International Journal of Occupational Safety and Ergonomics

ISSN: 1080-3548 (Print) 2376-9130 (Online) Journal homepage: www.tandfonline.com/journals/tose20

The input efficiency of chord keyboards

Fong-Gong Wu & Wen-Zhou Shi

To cite this article: Fong-Gong Wu & Wen-Zhou Shi (2018) The input efficiency of chord keyboards, International Journal of Occupational Safety and Ergonomics, 24:4, 638-645, DOI: 10.1080/10803548.2017.1362171

To link to this article: https://doi.org/10.1080/10803548.2017.1362171

Published online: 04 Sep 2017.

Submit your article to this journal

Article views: 191

View related articles

View Crossmark data

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tose20

International Journal of Occupational Safety and Ergonomics (JOSE), 2018
Vol. 24, No. 4, 638–645, https://doi.org/10.1080/10803548.2017.1362171

The input efficiency of chord keyboards

Fong-Gong Wu* and Wen-Zhou Shi

Department of Industrial Design, National Cheng Kung University, Taiwan

This study focused on an innovative mobile input device model that integrated ergonomic design principles to create a chord keyboard for use in concert with pointing devices. We conducted usability tests to demonstrate the operating performances of such keyboards using keys with four characters per key. Twelve right-handed subjects between the ages of 20 and 30 years and with no wrist disability participated in the usability tests. We analyzed the resulting data to compare the differences among the means. The results indicated that keyboards with multiple characters per key created some confusion among users but that keys with multiple characters per key nonetheless resulted in better input performance.

Keywords: chorded keyboard; pointing device; input efficiency

As technology has continued to develop, lifestyles have become increasingly influenced by this technology and dependent on the information products that have made our lives so convenient. The rapidly expanding use of mobile phone and computing technologies has led both to countless changes in the software and hardware components of such products and to intense interest in newly released products. The current environment, then, is particularly well suited to the development of innovative technological designs. In terms of keyboards, this environment of innovation should provide opportunities for keyboard design breakthroughs that rectify the deficiencies of current models.

1. Introduction

Current information technology (IT) devices are certainly more advanced and effective than earlier models. However, the ongoing development of new applications and the emergence of various broad technological trends (e.g., the aforementioned mobility trend) continuously reduce the utility of even the most advanced devices, as well as their capacity to satisfy our needs. Thus, there exists a constant need to imagine novel input devices, such as innovative keyboards, that could enhance the effectiveness of IT products. Efforts in this direction have led us to create new input operating models utilizing the chorded keyboard input method for use with pointing devices (such as a mouse or stylus) for mobile devices and computers.

According to Fitts' law, the time required to move an object (e.g., a finger or a computer cursor) to a target (such as a button on a keyboard or a link on a computer screen) is a function of the target's size and the distance of the object from it. The model has been mathematically formulated in various ways and has been widely adopted in numerous research areas, including kinematics, human factors and human–computer interaction (HCI). The chorded key concept employed in this study was developed based on Fitts' law with the aim being to lessen the moving distance required of fingers during keyboard operation and to significantly reduce the keyboard size, in addition to providing intuitive tactile, positioning and response characteristics for the fingers. Chord inputs can be made by pressing several keys at the same time, much as one plays a 'chord' on a piano by pressing several keys simultaneously; a well known example of a chord input would be the 'Ctrl–Alt–Delete' command used on IBM PC compatible computers. By placing multiple characters on individual keys, a keyboard using chorded keys can be substantially reduced in size and, thus, require users' fingers to travel shorter distances during input operations.

1.1. Research purpose

The present research considered possibilities beyond currently used keyboard models to propose new and innovative input operation methods while adhering to an ergonomic point of view. As already noted, Fitts' law was used to design chorded keys for use in keyboard designs that would reduce the moving distance required of fingers during keyboard operation while also significantly reducing keyboard size. The aim was to produce a keyboard input design that would be more efficient than current

*Corresponding author. Email: novafgw@gmail.com

© 2017 Central Institute for Labour Protection – National Research Institute (CIOP-PIB)

keyboards as the number of phone and computer products intended for mobile use continues to increase.

In short, the main objectives of this research were as follows:

- developing a chord input keyboard that will reduce the number of keys required and the size of the key groups;
- to reduce movement time by reducing the distance that the user's hand is required to move;
- to reduce the strain incurred by the human body due to keyboard usage by developing a keyboard design that allows for increased efficiency with less strain.

1.2. Literature review

1.2.1. The QWERTY keyboard

The design and key arrangement of the QWERTY keyboard (which was first patented in 1868 by Christopher Latham Sholes) were primarily based on an initial objective of reducing the use of consumables and the probability of malfunction in a typewriter. In consequence, many researchers subsequently criticized the shortcomings of the keyboard and proposed improvements [1], with the most famous example being the Dvorak keyboard proposed by Dvorak and his industrial engineering team in 1943 [2]. Regardless of the various suggestions put forward by Dvorak and other researchers, however, users remain used to the QWERTY keyboard layout, and hence there has been no alternative that has succeeded in replacing it.

In fact, while the small sizes of cell phones helped the MultiTap method (which requires fewer keys but also has the disadvantage of requiring increased key strokes per character) [3] to become the most commonly used method for inputting characters on cell phones, the more advanced technological features of smartphones and tablets (such as touchscreens) have allowed the QWERTY layout to be widely implemented in these devices.

At the same time, in order for a QWERTY keyboard to be used on a smartphone, the original 28-cm standard keyboard must be fitted onto a much smaller screen. Accordingly, the size of each individual key for touchscreen smartphones with miniature QWERTY keyboards is not well suited to the average human finger [4], a fact which clearly suggests that there is still room for improvement over currently popular QWERTY keyboard designs.

1.2.2. Chord inputting

Chord keyboards represent one of the more promising alternatives to the traditional QWERTY design because they allow users to make the same inputs using far fewer keys. For example, 5 keys on a chord keyboard can cover 31 different chord sets, representing characters, numbers, words, orders or a string of characters [1]. As already noted, the operating principle for chord keyboards is to simultaneously press several keys to input a single or multiple characters. Often, two or three keys need to be pressed together and, in fact, the chord concept is already used to a limited extent in current QWERTY keyboards: the Shift, Ctrl and Alt keys, e.g., are chord function keys. The advantages of fully realized chord keyboards lie primarily in the fact that they require less space for effective operation. Some specific advantages of chord keyboards are as follows [5]: (a) a chord keyboard has fewer keys and smaller dimensions than traditional keyboards, making it suitable for portable use; (b) certain chord keyboards support single-hand typing and are thus more user-friendly for disabled users; (c) chord keyboards can be more easily operated in moving or unstable conditions; (d) certain chord keyboards are resistant to damage due to repetitive force.

Compared to standard keyboards, chord keyboards are relatively slow in speed (i.e., the number of presses a user can make per minute) but perform better in terms of actually inputting information. In general, chord keyboards are difficult to master, especially their coding systems. However, input performance can reach a high level after a user becomes familiar with such a system. For example, Bau et al. [6] demonstrated that when users employ their Arpege technique for learning chord inputs on multi-touch devices, it is possible for those users to learn and retain a large vocabulary of chords. Furthermore, while chord keyboards that support single-hand operation were once widely seen among users with physical disabilities, they are now also increasingly popular among users who require quick input capabilities and portability. In addition, recent developments in chord keyboards have solved ergonomic problems and reduced keyboard dimensions, enhancing their feasibility for wider use.

The chord keyboard input keys used in this research must be complemented by a pointing device. The keyboards can be operated by a single hand, similar to the wrist personal computer keyboard, but the user may also wield the pointing device in the other hand simultaneously for some inputs. That said, for the purposes of inputs that involve pressing the keys themselves, the dimensions of the keys have been adjusted by referencing ergonomics studies to ensure that the finger can accurately press and hold the required keys. Gap distances between neighboring keys were also measured to reduce the chances of mistyping.

Research on earlier chord devices provided some guidance in designing the chord keyboards used in this study. For example, Twiddler™ (Tek Gear, Canada) is a chord device that allows single-hand inputting and has typically been used on mobile devices. Twiddler™ uses the ABC sequential arrangement and integrates key combinations

for chord inputs into the characters. Furthermore, it has a broad definition of input characters that allows many combinations to be defined as multiple characters (e.g., -ing, -tion) [4]. Despite the ability of Twiddler™ to improve character input efficiency, however, studies have pointed out that it entails a low floor entry rate, a long learning process and a high chance of error [7,8]. Moreover, Twiddler™ involves a high number of keys, which can affect efficiency for different users due to differences in gender, palm size and finger length [9]. Therefore, it is difficult to completely replicate Twiddler™ from a hand-held device to a cell phone. Thus, while cell phones are among the mobile devices on which Twiddler™ is commonly used, the efficiency of Twiddler™ is reduced when used on such devices. Accordingly, this research focused on the possibility of shortening the learning curve and effectively installing chord keys onto hand-held mobile devices.

A chord keyboard does not have many keys, resulting in overreliance on spatial correspondence or memory combinations. Therefore, it is also a non-visual or low-visual keyboard, as it does not rely on the sense of sight. A chord keyboard relies on good tactile feedback to achieve good input efficiency. On the other hand, users would need to spend considerable time and effort to learn the rules of chord inputting to improve efficiency and reduce errors. Sandnes and Aubert [10] believe that it is important for the user to get the hang of a device without the need for too much training. The new interactive methods thus have to coexist with existing capabilities in order not to overburden the user with new know-how [11].

With respect to ergonomic considerations, the present study used Fitts' law to shorten the distance traveled by the finger when operating the keyboard so as to increase efficiency when inputting text. In addition to Fitts' law, the transfer of learning is also critical because the prevalence and advantages of the QWERTY keyboard noted earlier, as well as its widespread usage, subtly affect users' general preferences and capabilities [10,12–15]. The effects of transfer of learning are made possible by procedural memory. Procedural memory is a special form of memory that is obtained through repetitive actions and does not require a high level of awareness from the user to operate once a certain action has effectively become a recorded procedure [15]. Therefore, experts who require quick typing can, when they have acquired the relevant procedural memory, locate the various character keys using their finger and without the need to search for them visually. As such, the design considerations for a new input method should include most users' familiarity with the QWERTY keyboard and the related procedural memory involving the relationships of the various fingers with different character keys. In contrast, an arrangement design that makes no allowances for the user's pre-established procedural memory could cause unnecessary interference [15].

A well-designed chord keyboard input method should allow users to input text using different key combinations in order to reduce the number of required keys. However, users have often criticized the memory requirements for such key combinations. The complexity entailed in such combinations has been pointed out by MacKenzie and Soukoreff [13], who stated that the higher the number of characters bundled into a single key, the greater the ambiguity that will be caused. When one single key corresponds to one single character, the ambiguity for the user is significantly reduced. However, a keyboard using scattered keys presents a steeper learning curve for the user than does a well-designed chord keyboard, the keys of which are easier to identify and remember due to their spatial correspondences. As such, an effective chord keyboard design must strike a balance between having too few characters per key and having too many, which is why the number of characters per key was a main variable focused on in this study.

At the same time, the current mobile IT devices rarely use actual physical keyboards; rather, for space and technical reasons, most devices, from smartphones to tablets, use simulated keyboards that users operate via touchscreens. As such, to be of practical use in future device designs, chord keyboard designs being researched should also consist of simulated keys manipulated via touchscreen. Leftheriotis et al. [16] recently demonstrated how large touchscreens can be modified into multi-touchscreens that allow multiple users to interact simultaneously to make chord inputs. In the present study, we applied the same concept to much smaller smartphone touchscreens, allowing a single user to make chord inputs by touching multiple keys at the same time.

1.3. Research methodologies

Applying ergonomics in the design of character keys can effectively reduce the damage caused to the human body during typing according to relevant research. Ergonomic principles for keyboards can be classified into the following five classes.

- Principles for operating between keyboard columns:
  - Achieve the highest possible main column-operating ratio: pressing rate of the main column should be above 55%.
  - Vertical movement across two columns should have the lowest possible percentage: vertical movement across more than two columns should be below 7%.
  - Movement within the same column should have the highest possible percentage: percentage of movement within the same column should be above 45%.
- Principles for operating with both hands:
  - The number of times for alternating of hands should be as high as possible: avoid using

one hand or one finger to continuously press the key.
  - The number of times for continuously using the same finger to press the key should be as low as possible: in terms of typing efficiency, operating by alternating hands > alternating fingers with a single hand > continuous operating with just a single finger [17].
  - Character keys with a high frequency to be linked together should be separated into the left and right sides, increasing the number of times for alternating of hands during operation.
- Principles for loading of both hands:
  - The ideal loading value for both hands is approximately 88.87/100, with the right hand operating load slightly higher than the left.
  - The loading rates of the various fingers (left little finger, left ring finger, left middle finger, left/right index finger, right middle finger, right ring finger and right little finger) are 9.8, 11.6, 12.8, 13.4, 14.4, 14.1, 12.5 and 11.3% respectively.
- Principles for clustering:
  - Klemmer and Stocker [18] conducted a study which found that, with respect to a seven-digit telephone number, recall time and accuracy will be better with clusters of three or four digits than for clusters of more or fewer digits. Therefore, the principle of clustering by three or four items should be used as a reference for the positioning of keys.
- Principles for arrangement:
  - Hirsch [19] proved that there is no significant difference in the rate of learning for an alphabetical keyboard whether the keys are arranged horizontally or vertically. An alphabetical keyboard can enable the user to find out the relative order

Table 1. Research on keyboard input operation and muscle stress.

Reference	Research purpose	Result of experiment	Research conclusion
[21]	To understand whether a new ergonomic keyboard can effectively reduce fatigue and musculoskeletal discomfort	Work efficiency is reduced when using the new keyboard	Use of different keyboards has no effect on fatigue
[22]	The effect on shoulder and hand muscle activity resulting from the positioning of an experimental mouse and keyboard	There is no significant relationship between the trapezius and the position of the mouse, whereas the level of deltoid muscle activity is significantly related to the position of the mouse	Female users would require more muscular activity on the right shoulder when using the mouse and keyboard
[23]	Confirmation of the damage done to the shoulder during usage of the mouse and keyboard	Muscle activity changes rapidly during usage of mouse and keyboard	The working posture is related to the rapid changes of individual muscle activity
[24]	Research on the differences in posture, muscle strength, cognitive ability, efficiency and gender when using the mouse and trackball for work	Female users would require more muscular activity on the right shoulder when using the mouse and keyboard	Recommends that the input device should be designed to be adjustable so as to adapt to different user dimensions
[25]	Evaluate how using a computer affects the muscles of different age groups	The EMG of seniors is higher than for young people, although there is only a small difference in precision action	The higher the frequency of using the computer, the higher the risk of injury to the shoulder skeletal muscle
[26]	Understand the effects on degree of muscle extension, wrist inclination angle and wrist and finger joints strength caused by different keyboard angles	A flat keyboard will lead to more extension	The more extended the wrist, the more inflexed the finger joints will be
[27]	Design of a keyboard that operates ideally	Keyboards with inclination and a separation design perform well	Different keyboard designs significantly affect operational performance and safety
[22]	Biomechanical evaluation of ergonomic keyboards	There is significant effect on the tendon when using an external keyboard	A movable keyboard can cause a maximum 11% effect on the tendon
[28]	Deduce the existing relationship between using the mouse and musculoskeletal symptoms	Extension of arms, insufficient screen height, nervousness and an overly raised shoulder can easily cause neck symptoms	With respect to damage due to computer usage, there is an increase in percentage of hazardous elements due to usage of the mouse

Note: EMG = electromyographic.

of positioning of the characters using a memory search before locating its position on the keyboard using a visual search. This combination of memory search and visual search is less effective as compared to purely locating the keys visually [19]. Considering the clustering principle and the fact that it is easier to search for, the horizontal arrangement principle was also considered for keyboard design.

You [20] discussed keyboard dimensions, forward inclination angle, wrist support configuration and their effects on typing efficiency, fatigue, subjective assessment and working posture. After conducting research using eight types of keyboards made up from these three sets of variables, it was observed that keyboard dimensions have a significant effect on typing performance as a wider keyboard leads to a faster typing speed; in terms of the chances of error, there is no significant effect. Furthermore, it can be seen that keyboard dimensions will be affected by the finger and wrist width of the user. A wider keyboard will be chosen by a subject user with a larger finger and wrist width. The forward inclination angle has an effect on wrist joints. When the keyboard forward inclination angle is 12.8°, the wrist deviation and extension angle will be greater and subject users tend to prefer the wider keyboard. Table 1 summarizes a number of research findings relating to keyboard input operation and the resulting stress to muscles [21–29].

2. Materials and methods

2.1. Keyboard design

The novel chord keyboard created for this study was ergonomically designed to effectively reduce strain caused to the user while increasing input efficiency. The keyboard had five keys, each of which was separated into four quadrants; two characters were designated to each quadrant. The keyboard was designed to be used in tandem with either a mouse or a pointing device. The button on the side of the mouse or stylus was used to shift between the two characters in each quadrant (Figure 1). The key placement and chord combinations themselves were chosen based on their common usage and previous research.

2.2. Participants

The 12 test subjects consisted of 6 males and 6 females, all of whom had at least 1 year of experience with IT product inputs. The English typing ability of each of the subjects ranged from 20 to 30 words per minute (wpm). The subjects were right-handed without any wrist disability, were physically and mentally healthy and were aged between 20 and 30 years.

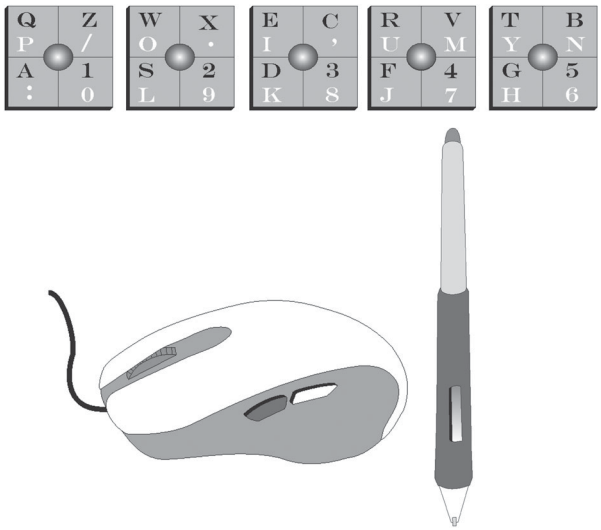

Figure 1. Design for a keyboard featuring chorded keys.

2.3. Experimental parameters

The primary independent variable in this study was to determine the relative efficiency of the chord keyboard layout in terms of input speed. The dependent variables included the input rate and the error rate. The input rate was defined as the number of characters entered according to the articles provided by the researchers within the experiment time (6 min). The error rate was defined as the number of incorrectly entered characters divided by the number of characters entered.

Participants were provided with the same chord keyboard and pointing device. Four English articles were adapted (from www.discovery.com) for the participants to retype. Environmental factors including lighting and physical setup were consistent for all subjects.

2.4. Experimental process and procedures

Input performance data for the chord keyboard were collected, including the time spent to input each character and the number of incorrect characters entered by the test participants. After the experiment was completed, the participants completed questionnaires regarding their opinion of the keyboard. First, participants were given a code table to help understand each chord arrangement and 5 min to familiarize themselves with the keyboard layout and input method. The participants were then provided with an article and allowed 6 min to type as much as possible. After the test, participants were given 3 min to rest before they were asked to type a second article using the other pointing device. The order of pointing devices tested was randomized to ensure internal validity.

The keyboard was placed on a table and was equipped with an LCD screen that displayed the input content (Figure 2). The same experimental procedures were

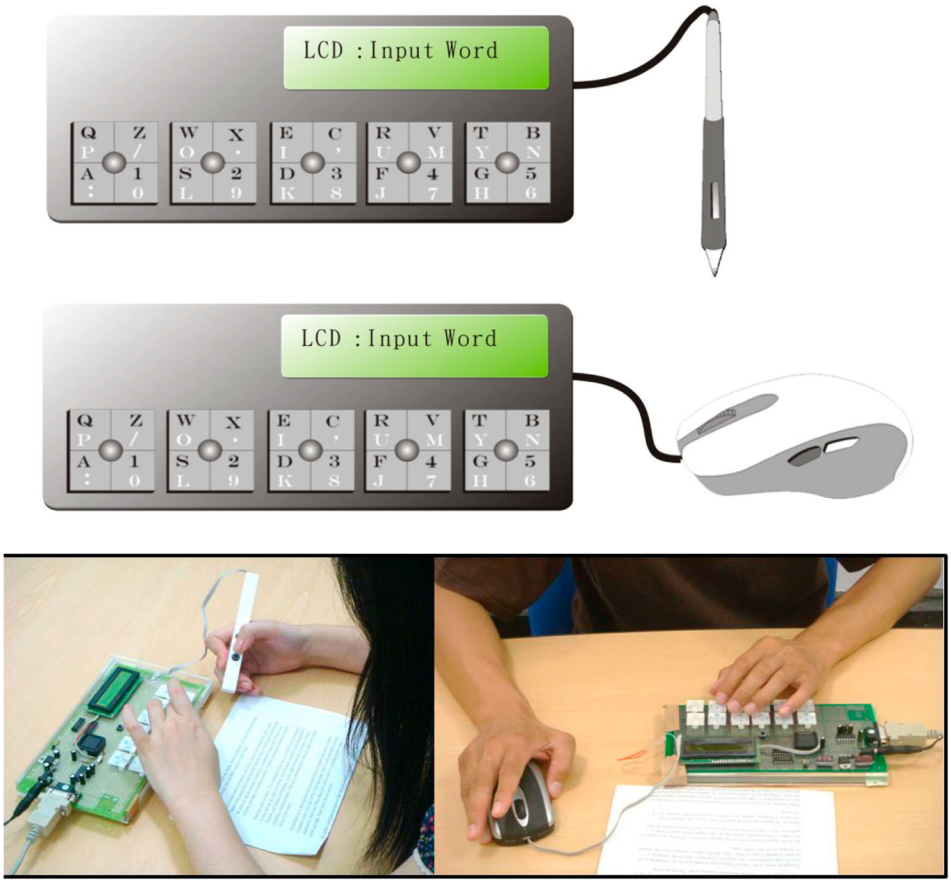

Figure 2. Experimental setup.

performed each day for 3 consecutive days and completed a subjective questionnaire at the end of the third day.

3. Results and discussion

3.1. *Analysis of text input rates*

In order to determine whether participants' input speeds improved over the 3 days, paired-sample *t* tests were completed on the speeds from days 1 and 3. The results are presented in Table 2. Input speeds on day 3 showed a significant improvement over those on day 1 for both the mouse and stylus groups, indicating the participants could input text faster with practice. Although input speeds increased each day, the amount of improvement will inevitably decrease over time.

3.2. *Analysis of error rates*

In order to determine whether participants' input error rate improved over the 3 days, paired-sample *t* tests were completed on the error rates from days 1 and 3. The results are presented in Table 3. Errors made on day 3 showed a significant improvement over those on day 1 for both the mouse and stylus groups, indicating that the participants could input text more accurately with practice. Although input errors decreased each day, the amount of improvement will inevitably decrease over time.

3.3. *Participant questionnaires*

Participants filled out questionnaires at the end of the final experiment. Participants were asked to rate the

Table 2. Paired-sample *t*-test results for chord keyboard input speed.

Keyboard type	Paired-sample *t*-test differences			*t*	*df*	Significance (two-tailed)
	M	*SD*	95% CI			
Chord keyboard with stylus	(−)66.9	22.69	[−83.13, −50.67]	(−)9.32	9	0.000*
Chord keyboard with mouse	(−)76.2	35.53	[−101.61, −50.79]	(−)6.78	9	0.000*

*Significance defined as $p < 0.05$.
Note: CI = confidence interval.

Table 3. Paired-sample *t*-test results for chord keyboard input errors.

Keyboard type	Paired-sample *t*-test differences			*t*	*df*	Significance (two-tailed)
	M	*SD*	95% CI			
Chord keyboard with stylus	1.216	0.47	[0.88, *e*]	*e*	9	0.000*
Chord keyboard with mouse	2.073	2.11	[0.56, 3.59]	3.10	9	0.013*

*Significance defined as $p < 0.05$.
Note: CI = confidence interval; e = mathematical constant e.

ease of learnability of the chord keyboard as well as their level of hand and wrist comfort on scales from 1 = *easy*/*comfortable* to 10 = *difficult*/*painful*. The average ease of learnability score for the chord keyboard with stylus was 7.84 and the average ease of learnability score for the chord keyboard with mouse was 7.38. This indicates that the different pairings affected participants' ability to adapt to this new keyboard. Left-hand comfort scored an average of 6.07; participants expressed that the keyboard was more comfortable than traditional keyboards as it took up less space and contained less keys. Right-hand comfort scored higher for the stylus than for the mouse with an average of 5.63, indicating the different peripherals affect hand comfort.

4. Conclusions and recommendations

Rapid technological advancements have brought computers and electronics into nearly every aspect of life. As we become more reliant on technology, communications products become more essential in our daily lives, especially mobile electronics such as laptops, tablet computers and smartphones. As noted in this study, inputting text on traditional QWERTY keyboards using two hands is more efficient than using a single hand. As such, the development of the novel keyboard in this study used chorded arrangements along with a pointing device in an attempt to provide a more portable, comfortable and efficient alternative.

This study showed that input efficiency using the chord keyboard along with both pointing devices increased along with practice. Input speed increased significantly and input errors decreased significantly over the course of this study. The use of a stylus instead of a mouse may effectively reduce hand and wrist strain. The pairing of a chord keyboard and stylus is especially applicable in mobile computing devices, in particular, tablet computers. The model used in this study can serve as a reference for future keyboards that may be used as peripherals for tablets and smartphones.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial or not-for-profit sectors.

References

[1] Noyes J. Chord keyboards. Appl Ergon. 1983;14:55–59. doi:10.1016/0003-6870(83)90221-1
[2] Dvorak A. There is a better typewriter keyboard. Natl Bus Edn Q. 1943;12:51–58, 66.
[3] MacKenzie IS. KSPC (keystrokes per character) as a characteristic of text entry techniques. Proceedings of the 4th International Symposium on Mobile Human–Computer Interaction; 2002 Sep 18–20; Pisa, Italy. Berlin: Springer; 2002. p. 195–210.
[4] Wigdor DJ. Chording and tilting for rapid, unambiguous text entry to mobile phones [master's thesis]. Toronto (ON): University of Toronto; 2004.
[5] Buxton W. Human input to computer systems: theories, techniques and technology. Cambridge (UK): Cambridge University Press; 2002. Chapter 6, Case study 2: chord keyboards.
[6] Bau O, Ghomi E, Mackay W. Arpege: design and learning of multi-finger chord gestures. LRI; 2010. (Technical report; no. 1533).
[7] Kristensson PO. Discrete and continuous shape writing for text entry and control [dissertation]. Linköping: Linköping University; 2007.
[8] Tinwala H. Eyes-free text entry on mobile devices [master's thesis]. Toronto (ON): York University; 2009.
[9] Lyons K, Starner T, Plaisted D, et al. Twiddler typing: one-handed chording text entry for mobile phones. Proceedings of the SIGCHI Conference on Human Factors in Computing Systems; 2004 Apr 24–29; Vienna, Austria. New York (NY): ACM; 2004. p. 671–678.
[10] Sandnes FE, Aubert A. Bimanual text entry using game controllers: relying on users' spatial familiarity with QWERTY. Interact Comput. 2007;19:140–150. doi:10.1016/j.intcom.2006.08.003
[11] Sandnes FE, Jian HL. Pair-wise variability index: evaluating the cognitive difficulty of using mobile text entry systems. Proceedings of the Mobile Human–Computer Interaction – Mobile HCI 2004; 2004 Sep 13–16; Glasgow, UK. Berlin: Springer; 2004. p. 347–350.
[12] Hwang S, Lee G. Qwerty-like 3×4 keypad layouts for mobile phone. Proceedings of the CHI '05 Extended Abstracts on Human Factors in Computing Systems; 2005 Apr 2–7; Portland, OR, USA. New York (NY): ACM; 2005. p. 1479–1482.
[13] MacKenzie IS, Soukoreff RW. Text entry for mobile computing: models and methods, theory and practice. Hum Comput Int. 2002;17:147–198. doi:10.1207/S15327051HCI172&3_2

[14] Matias E, MacKenzie IS, Buxton W. One-handed touch typing on a QWERTY keyboard. Hum Comput Int. 1996; 11:1–27. doi:10.1207/s15327051hci1101_1
[15] McLoone H, Hinckley K, Cutrell E. Bimanual interaction on the Microsoft Office keyboard. In: Rauterberg GWM, Menozzi M, Wesson J, editors. Human–computer interaction INTERACT '03. Amsterdam: IOS Press; 2003. p. 49–56.
[16] Leftheriotis I, Chorianopoulos K, Jaccheri L. Tool support for developing scalable multiuser applications on multitouch screens. Proceedings of the 2012 ACM International Conference on Interactive Tabletops and Surfaces; 2012 Nov 11–14; Cambridge/Boston, MA, USA: ACM. p. 371–374.
[17] Kinkead R. Typing speed, keying rates, and optimal keyboard layouts. Proc Hum Factors Ergon Soc Annu Meet. 1975;19(2):159–161. doi:10.1177/154193127501900203
[18] Klemmer ET, Stocker LP. Effects of grouping of printed digits on forced-paced manual entry performance. J Appl Psych. 1974;59:675–678. doi:10.1037/h0037513
[19] Hirsch J, inventor; Hirsch J, assignee. Tactile control system. United States patent US 3,497,668. 1970 Feb 24.
[20] You ZW. Effects of keyboard size, frontal inclination and wrist rest on keying performance, subjective fatigue and operating posture [master's thesis]. Taipei: National Taiwan University of Science and Technology; 2000.
[21] Swanson NG, Galinsky TL, Cole LL, et al. The impact of keyboard design on comfort and productivity in a text-entry task. Appl Ergon. 1997;28(1):9–16. doi:10.1016/S0003-6870(96)00052-X
[22] Treaster DE, Marras WS. An assessment of alternate keyboards using finger motion, wrist motion and tendon travel. Clin Biomech (Bristol, Avon). 2000;15(7):499–503. doi:10.1016/S0268-0033(00)00012-7
[23] Cooper A, Straker L. Mouse versus keyboard use: a comparison of shoulder muscle load. Int J Ind Ergon. 1998;22(4):351–357. doi:10.1016/S0169-8141(97)00088-7
[24] Karlqvist L, Bernmark E, Ekenvall L, et al. Computer mouse and track-ball operation: similarities and differences in posture, muscular load and perceived exertion. Int J Ind Ergon. 1999;23:157–169. doi:10.1016/S0169-8141(97)00031-0
[25] Laursen B, Jensen BR. Shoulder muscle activity in young and older people during a computer mouse task. Clin Biomech (Bristol, Avon). 2000;15:S30–S33. doi:10.1016/S0268-0033(00)00057-7
[26] Nelson JE, Treaster DE, Marras WS. Finger motion, wrist motion and tendon travel as a function of keyboard angles. Clin Biomech (Bristol, Avon). 2000;15(7):489–498. doi:10.1016/S0268-0033(00)00011-5
[27] Gilad I, Harel S. Muscular effort in four keyboard designs. Int J Ind Ergon. 2000;26(1):1–7. doi:10.1016/S0169-8141(99)00057-8
[28] Cook C, Burgess-Limerick R, Chang S. The prevalence of neck and upper extremity musculoskeletal symptoms in computer mouse users. Int J Ind Ergon. 2000;26(3):347–356. doi:10.1016/S0169-8141(00)00010-X
[29] Cook CJ, Kothiyal K. Influence of mouse position on muscular activity in the neck, shoulder and arm in computer users. Appl Ergon. 1998;29(6):439–443. doi:10.1016/S0003-6870(98)00008-8